Foundation Certificate in Higher Education

Module: DOC333 – Introduction to Programming Principles

Module Leader: Mr. Welihinda

Assignment Number: 1

Type of Assignment: Individual Coursework

Submission Date: 19th, July 2024

Branch: Colombo

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Abstract

The purpose of this project is to develop an information system for ABC Bank in Sri Lanka that efficiently manages the details of their savings account holders. The system will streamline the process of opening new accounts, facilitate the updating of customer, and enable the bank to manage deposits and withdrawals. Additionally, the system will provide functionalities to display all customers with their account numbers, names, and balances, as well as view the details of individual customers, thereby enhancing customer service and operational efficiency.

Case

ABC bank is a small bank in Sri Lanka to register their customers to serve them fast. The bank is planning to have an information system to maintain the details of their savings account holders. The system must be able to facilitate opening accounts to new customers and display all the customers in the bank with their account number, name and the bank balance when needed. Also, the system should have the facility to change details of the customers already registered with the bank, and to deposit and withdraw money when required for a given account. The bank should also be able to view the details of a customer for a given account when required.

Algorithm

1) Start
2) Display the main menu.
3) Get an input from the user.
   a) If the user entered 1:
      i. Redirect the user to the “Add a new customer.”
      ii. Get the relevant information from the user.
      iii. Prompt the user to decide whether they want to save it or not.
   b) If the user entered 2:
      i. Redirect the user to the “View details of a customer including his/her bank balance.”
      ii. Ask the user to input the account number.
      iii. Prompt the user to decide whether they want to view details of another account or not.
   c) If the user entered 3:
      i. Redirect the user to the “View details of all the customers including their bank balances.”
      ii. Prompt the user to decide whether they want to update any customer details.
   d) If the user entered 4:
      i. Redirect the user to the “Deposit money to a given account.”
      ii. Ask the user to input the account number.
      iii. Ask the user to enter the amount they wish to deposit.
      iv. Prompt the user to decide whether they want to save it or not.
   e) If the user entered 5:
      i. Redirect the user to the “Withdraw money from a given account.”
      ii. Ask the user to enter the amount they wish to withdraw.
      iii. Prompt the user to decide whether they want to save it or not.
   f) If the user entered 6:
      i. Redirect the user to the “Update Customer Details.”
      ii. Ask the user to input the account number.
      iii. Add and update the criteria which needed an update.
   g) If the user entered 7:
      i. Go to step 5.
4) Return to step 2.
5) Exit

Special Assumptions

1. The default number of customers is zero.
2. The birthdate is expected to be in the ISO 8601 format, or YYYY-MM-DD.
3. The maximum amount that can be deposited at once is 5,000,000.
4. The minimum amount that can be withdrawn is 100.
5. The maximum amount that can be withdrawn once is 60,000.

Added Features

1. If the user does not respond to the prompt asking whether they want to save their information, it will be discarded.

Test Cases

Positive Cases

1. Main Menu

Test Case	Input	Expected Outcome	Actual Outcome	Result
1	-	1. Add a new customer	1. Add a new customer	Pass
	-	2. View details of a customer including his/her bank balance	2. View details of a customer including his/her bank balance	Pass
	-	3. View details of all the customers with their bank balances	3. View details of all the customers with their bank balances	Pass
	-	4. Deposit money to a given account	4. Deposit money to a given account	Pass
	-	5. Withdraw money from a given account	5. Withdraw money from a given account	Pass
	-	6. Update Customer Details	6. Update Customer Details	Pass
	-	7. Exit	7. Exit	Pass
	Expected from the user	Your Choice:	Your Choice:	Pass

Table 1: Test Case 1 (Main Menu)

```
*IDLE Shell 3.12.4*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun  6 2024, 19:30:16) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\User\Downloads\Nadithi cw.py

ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: |
```

Figure 1:Test Case 1 (Main Menu)

1.1. Add New Customer

Test Case	Input	Assumed output	Actual output	Result
2	1812200625	Bank Account Number: 1812200625	Bank Account Number: 1812200625	Pass
	2005630317	NIC: 2005630317	NIC: 2005630317	Pass
	Grace	First Name: Grace	First Name: Grace	Pass
	William	Last Name: William	Last Name: William	Pass
	2005-11-18	Birth Date: 2005-11-18	Birth Date: 2005-11-18	Pass
	49 Station Road	Permanent Address: 49 Station Road	Permanent Address: 49 Station Road	Pass

	+447911123	Phone Number: +447911123	Phone Number: +447911123	Pass
	yes	Do you want to save this new customer? (yes/no): yes	Do you want to save this new customer? (yes/no): yes	Pass
	-	Customer added successfully!	Customer added successfully!	Pass

Table 2: Test Case 2 (Add a new customer)

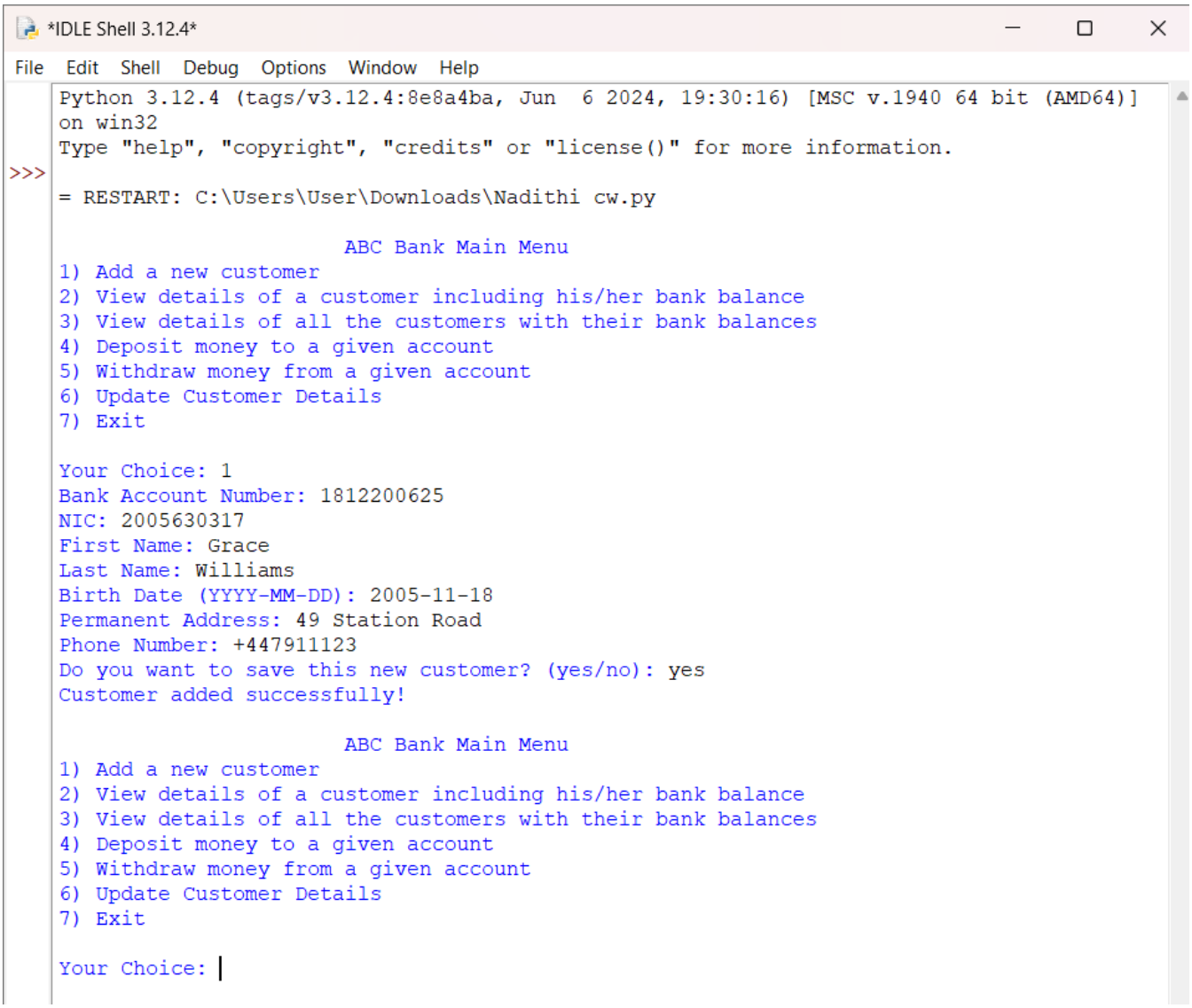
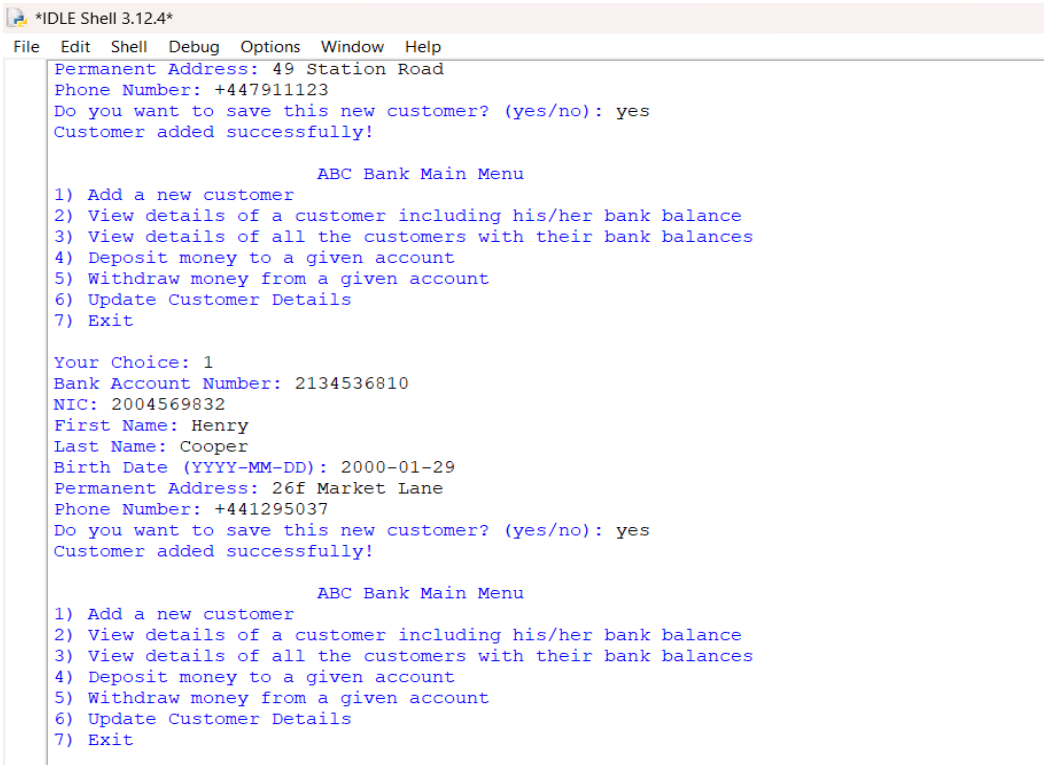


Figure 2: Test Case 2 (Add a new customer)

Test Case	Input	Assumed output	Actual output	Result
3	2134536810	Bank Account Number: 2134536810	Bank Account Number: 2134536810	Pass
	2004569832	NIC: 2004569832	NIC: 2004569832	Pass
	Henry	First Name: Henry	First Name: Henry	Pass
	Cooper	Last Name: Cooper	Last Name: Cooper	Pass
	2000-01-29	Birth Date: 2000-01-29	Birth Date: 2000-01-29	Pass
	26f Market Lane	Permanent Address: 26f Market Lane	Permanent Address: 26f Market Lane	Pass
	+441295037	Phone Number: +441295037	Phone Number: +441295037	Pass
	yes	Do you want to save this new customer? (yes/no): yes	Do you want to save this new customer? (yes/no): yes	Pass
	-	Customer added successfully!	Customer added successfully!	Pass

Table 3: Test Case 2 (Add a new customer)



```
*IDLE Shell 3.12.4*
File  Edit  Shell  Debug  Options  Window  Help
Permanent Address: 49 Station Road
Phone Number: +447911123
Do you want to save this new customer? (yes/no): yes
Customer added successfully!

                    ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 1
Bank Account Number: 2134536810
NIC: 2004569832
First Name: Henry
Last Name: Cooper
Birth Date (YYYY-MM-DD): 2000-01-29
Permanent Address: 26f Market Lane
Phone Number: +441295037
Do you want to save this new customer? (yes/no): yes
Customer added successfully!

                    ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit
```

Figure 3: Test Case 2 (Add a new customer)

Test Case	Input	Assumed output	Actual output	Result
4	8519374621	Bank Account Number: 8519374621	Bank Account Number: 8519374621	Pass
	2006755691	NIC: 2006755691	NIC: 2006755691	Pass
	Tom	First Name: Tom	First Name: Tom	Pass
	Hanks	Last Name: Hanks	Last Name: Hanks	Pass
	1990-04-15	Birth Date: 1990-04-15	Birth Date: 1990-04-15	Pass
	10 Downing St	Permanent Address: 10 Downing St	Permanent Address: 10 Downing St	Pass
	+449751248	Phone Number: +449751248	Phone Number: +449751248	Pass
	yes	Do you want to save this new customer? (yes/no): yes	Do you want to save this new customer? (yes/no): yes	Pass
	-	Customer added successfully!	Customer added successfully!	Pass

Table 4: Test Case 2 (Add a new customer)

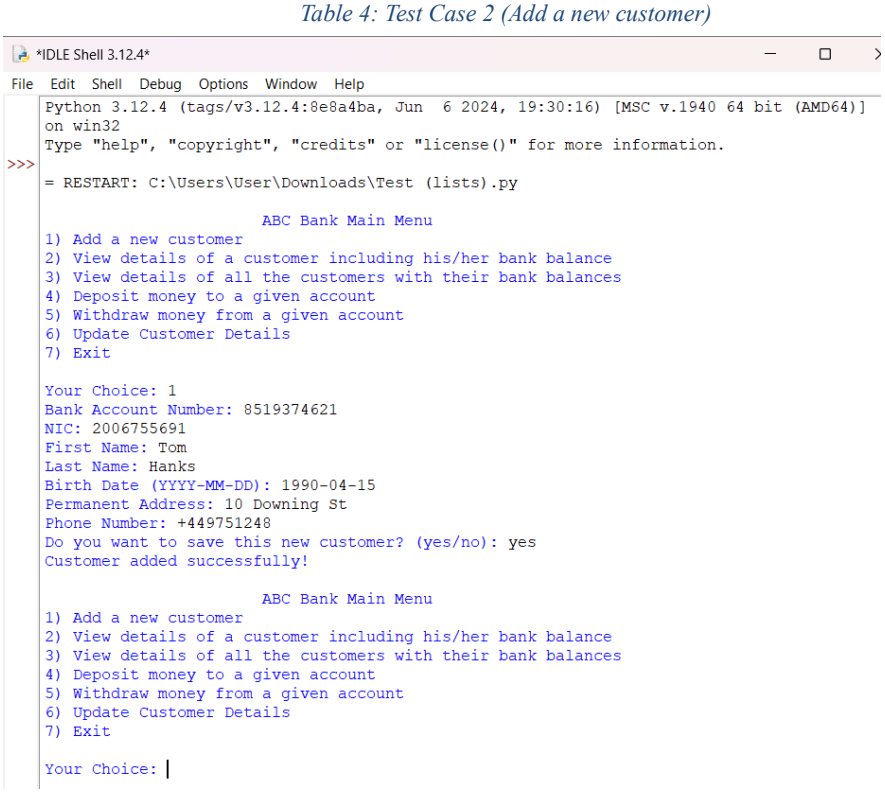
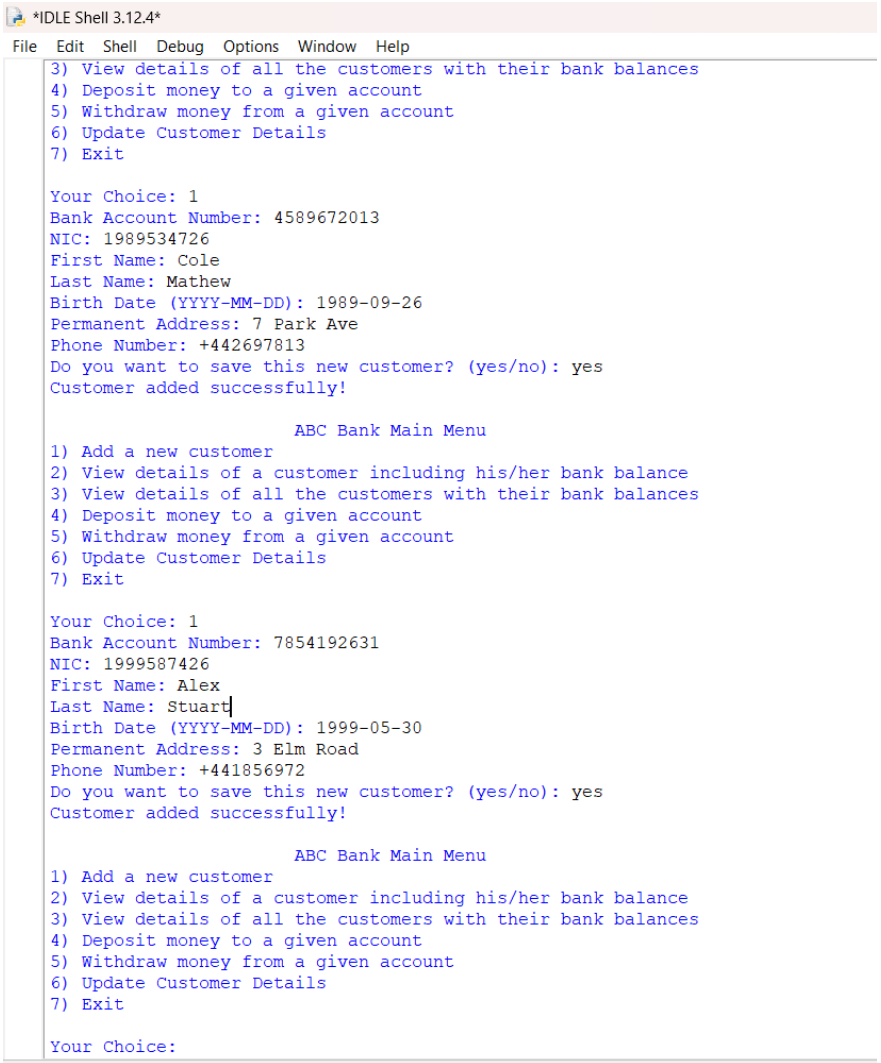


Figure 4: Test Case 2 (Add a new customer)

Test Case	Input	Assumed output	Actual output	Result
5	4589672013	Bank Account Number: 4589672013	Bank Account Number: 4589672013	Pass
	1989534726	NIC: 1989534726	NIC: 1989534726	Pass
	Cole	First Name: Cole	First Name: Cole	Pass
	Mathew	Last Name: Mathew	Last Name: Mathew	Pass
	1989-09-26	Birth Date: 1989-09-26	Birth Date: 1989-09-26	Pass
	7 Park Ave	Permanent Address: 7 Park Ave	Permanent Address: 7 Park Ave	Pass
	+442697813	Phone Number: +442697813	Phone Number: +442697813	Pass
	yes	Do you want to save this new customer? (yes/no): yes	Do you want to save this new customer? (yes/no): yes	Pass
	-	Customer added successfully!	Customer added successfully!	Pass

Table 5: Test Case 2 (Add a new customer)



```
*IDLE Shell 3.12.4*
File  Edit  Shell  Debug  Options  Window  Help
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 1
Bank Account Number: 4589672013
NIC: 1989534726
First Name: Cole
Last Name: Mathew
Birth Date (YYYY-MM-DD): 1989-09-26
Permanent Address: 7 Park Ave
Phone Number: +442697813
Do you want to save this new customer? (yes/no): yes
Customer added successfully!

                    ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 1
Bank Account Number: 7854192631
NIC: 1999587426
First Name: Alex
Last Name: Stuart
Birth Date (YYYY-MM-DD): 1999-05-30
Permanent Address: 3 Elm Road
Phone Number: +441856972
Do you want to save this new customer? (yes/no): yes
Customer added successfully!

                    ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice:
```

Figure 5:Test Case 2 (Add a new customer)

Test Case	Input	Assumed output	Actual output	Result
6	7854192631	Bank Account Number: 7854192631	Bank Account Number: 7854192631	Pass
	1999587426	NIC: 1999587426	NIC: 1999587426	Pass
	Alex	First Name: Alex	First Name: Alex	Pass
	Stuart	Last Name: Stuart	Last Name: Stuart	Pass
	1999-05-30	Birth Date: 1999-05-30	Birth Date: 1999-05-30	Pass

	3 Elm Road	Permanent Address: 3 Elm Road	Permanent Address: 3 Elm Road	Pass
	+441856972	Phone Number: +441856972	Phone Number: +441856972	Pass
	yes	Do you want to save this new customer? (yes/no): yes	Do you want to save this new customer? (yes/no): yes	Pass
	-	Customer added successfully!	Customer added successfully!	Pass

Table 6: Test Case 2 (Add a new customer)

1.1.1. If the customer input "no"

Test Case	Input	Assumed output	Actual output	Result
7	1812200625	Bank Account Number: 1812200625	Bank Account Number: 1812200625	Pass
	2005630317	NIC: 2005630317	NIC: 2005630317	Pass
	Grace	First Name: Grace	First Name: Grace	Pass
	William	Last Name: Williams	Last Name: Williams	Pass
	2005-11-18	Birth Date: 2005-11-18	Birth Date: 2005-11-18	Pass
	49 Station Road	Permanent Address: 49 Station Road	Permanent Address: 49 Station Road	Pass
	+447911123	Phone Number: +447911123	Phone Number: +447911123	Pass
	no	Do you want to save this new customer? (yes/no): no	Do you want to save this new customer? (yes/no): no	Pass
	-	Customer not saved.	Customer not saved.	Pass

Table 7: Test Case 3 (Add new customer-If the customer input "no")

```
*IDLE Shell 3.12.4*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun  6 2024, 19:30:16) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\User\Downloads\Test (lists).py

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 1
Bank Account Number: 1812200625
NIC: 2005630317
First Name: Grace
Last Name: William
Birth Date (YYYY-MM-DD): 2005-11-18
Permanent Address: 49 Station Road
Phone Number: +447911123
Do you want to save this new customer? (yes/no): no
Customer not saved.

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice:
```

Figure 6: Test Case 3 (Add new customer-If the customer input "no")

1.2 View details of a customer including his/her bank balance

Test Case	Input	Assumed Output	Actual Output	Results
8	1812200625	Enter Account Number: 181220025	Enter Account Number: 181220025	pass
		Account Number: 181220025	Account Number: 181220025	Pass
		NIC: 2005630317	NIC: 2005630317	Pass
		Name: Grace William	Name: Grace William	pass
		Balance: 0	Balance:0	Pass
	Yes	Do you want to view the details of another account? (yes/no): yes	Do you want to view the details of another account? (yes/no): yes	Pass
	2134536810	Enter account number: 2134536810	Enter account number: 2134536810	Pass

Table 8: Test Case 4 (View details of a customer including his/her bank balance)

```
*IDLE Shell 3.12.4*
File  Edit  Shell  Debug  Options  Window  Help
Permanent Address: 49 Station Road
Phone Number: +447911123
Do you want to save this new customer? (yes/no): yes
Customer added successfully!

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 1
Bank Account Number: 2134536810
NIC: 2004569832
First Name: Henry
Last Name: Cooper
Birth Date (YYYY-MM-DD): 2000-01-29
Permanent Address: 26f Market Lane
Phone Number: +441295037
Do you want to save this new customer? (yes/no): yes
Customer added successfully!

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 2
Enter account number: 1812200625
Account Number: 1812200625
NIC: 2005630317
Name: Grace William
Balance: 0
Do you want to view details of another account? (yes/no): yes
Enter account number: 2134536810
Account Number: 2134536810
NIC: 2004569832
Name: Henry Cooper
Balance: 0
Do you want to view details of another account? (yes/no):
```

Figure 7: Test Case 4 (View details of a customer including his/her bank balance)

1.2.1 If the user input "no"

Test Case	Input	Assumed Output	Actual Output	Results
9	1812200625	Enter Account Number: 181220025	Enter Account Number: 181220025	pass
		Account Number: 181220025	Account Number: 181220025	Pass

		NIC: 2005630317	NIC: 2005630317	Pass
		Name: Grace William	Name: Grace William	pass
		Balance: 0	Balance:0	Pass
	no	Do you want to view the details of another account? (yes/no): no	Do you want to view the details of another account? (yes/no): no	Pass

Table 9: Test Case 5 (View details of a customer including his/her bank balance- If the user input "no")

```
*IDLE Shell 3.12.4*
File  Edit  Shell  Debug  Options  Window  Help
                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 1
Bank Account Number: 1812200625
NIC: 2005630317
First Name: Grace
Last Name: William
Birth Date (YYYY-MM-DD): 2005-11-18
Permanent Address: 49 Station Road
Phone Number: +447911123
Do you want to save this new customer? (yes/no): yes
Customer added successfully!

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 2
Enter account number: 1812200625
Account Number: 1812200625
NIC: 2005630317
Name: Grace William
Balance: 0
Do you want to view details of another account? (yes/no): no

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: |
```

Figure 8: Test Case 5 (View details of a customer including his/her bank balance-If the user input "no")

1.3 View details of all the customers with their bank balances

Test Case	Input	Assumed Output	Actual Output	Results
10	-	Account Number: 181220025	Account Number: 181220025	pass
	-	NIC: 2005630317	NIC: 2005630317	Pass
	-	Name: Grace William	Name: Grace William	pass
	-	Balance: 0	Balance:0	Pass
	yes	Do you want to update any customer details? (yes/no): yes	Do you want to update any customer details? (yes/no): yes	Pass
	181220025	Enter account number: 181220025	Enter account number: 181220025	Pass

Table 10: Test Case 6 (View details of all the customers with their bank balance)

```
*IDLE Shell 3.12.4*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun  6 2024, 19:30:16) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\User\Downloads\Test (lists).py

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 1
Bank Account Number: 1812200625
NIC: 2005630317
First Name: Grace
Last Name: William
Birth Date (YYYY-MM-DD): 2005-11-18
Permanent Address: 49 Station Road
Phone Number: +447911123
Do you want to save this new customer? (yes/no): yes
Customer added successfully!

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 3
Account Number: 1812200625
NIC: 2005630317
Name: Grace William
Balance: 0
Do you want to update any customer details? (yes/no): yes
Enter account number:
```

Figure 9: Test Case 6 (View details of all the customers with their bank balance)

1.3.1 If user inputs “no”

Test Case	Input	Assumed Output	Actual Output	Results
11	-	Account Number: 181220025	Account Number: 181220025	pass
	-	NIC: 2005630317	NIC: 2005630317	Pass
	-	Name: Grace William	Name: Grace William	pass
	-	Balance: 0	Balance:0	Pass
	no	Do you want to update any customer details? (yes/no): no	Do you want to update any customer details? (yes/no): no	Pass

Table 11:Test Case 7 (View details of all the customers including bank balance- If the user inputs "no")

```
*IDLE Shell 3.12.4*
File  Edit  Shell  Debug  Options  Window  Help
                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 1
Bank Account Number: 1812200625
NIC: 2005630317
First Name: Grace
Last Name: William
Birth Date (YYYY-MM-DD): 2005-11-18
Permanent Address: 49 Station Road
Phone Number: +447911123
Do you want to save this new customer? (yes/no): yes
Customer added successfully!

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 3
Account Number: 1812200625
NIC: 2005630317
Name: Grace William
Balance: 0
Do you want to update any customer details? (yes/no): no

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: |
```

Figure 10: Test Case 7 (View details of all the customers including bank balance- If the user inputs "no")

1.4 Deposit money to a given account.

Test Case	Input	Assumed Output	Actual Output	Results
12	1812200625	Enter Account Number: 181220025	Enter Account Number: 181220025	pass
	200000	Enter amount to deposit: 200000	Enter amount to deposit: 200000	Pass
	yes	Do you want to save? (yes/no): yes	Do you want to save? (yes/no): yes	Pass
	-	Deposit successful!	Deposit successful!	Pass
	-	New balance: 200000.00	New balance: 200000.00	Pass

Table 12: Test Case 8 (Deposit money to a given account)

```
*IDLE Shell 3.12.4*
File  Edit  Shell  Debug  Options  Window  Help
6) Update Customer Details
7) Exit

Your Choice: 1
Bank Account Number: 1812200625
NIC: 2005630317
First Name: Grace
Last Name: William
Birth Date (YYYY-MM-DD): 2005-11-18
Permanent Address: 49 Station Road
Phone Number: +447911123
Do you want to save this new customer? (yes/no): yes
Customer added successfully!

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 4
Enter account number: 1812200625
Enter amount to deposit: 200000
Do you want to save? (yes/no): yes
Deposit successful!
New balance: 200000.00

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice:
```

Figure 11: Test Case 8 (Deposit money to a given account)

1.4.1 If the user inputs “no”

Test Case	Input	Assumed Output	Actual Output	Results
13	1812200625	Enter Account Number: 181220025	Enter Account Number: 181220025	pass
	200000	Enter amount to deposit: 200000	Enter amount to deposit: 200000	Pass
	no	Do you want to save? (yes/no): no	Do you want to save? (yes/no): no	Pass
	-	Deposit canceled.	Deposit canceled.	Pass

Table 13: Test Case 9 (Deposit money to a given account-If the user inputs "no")

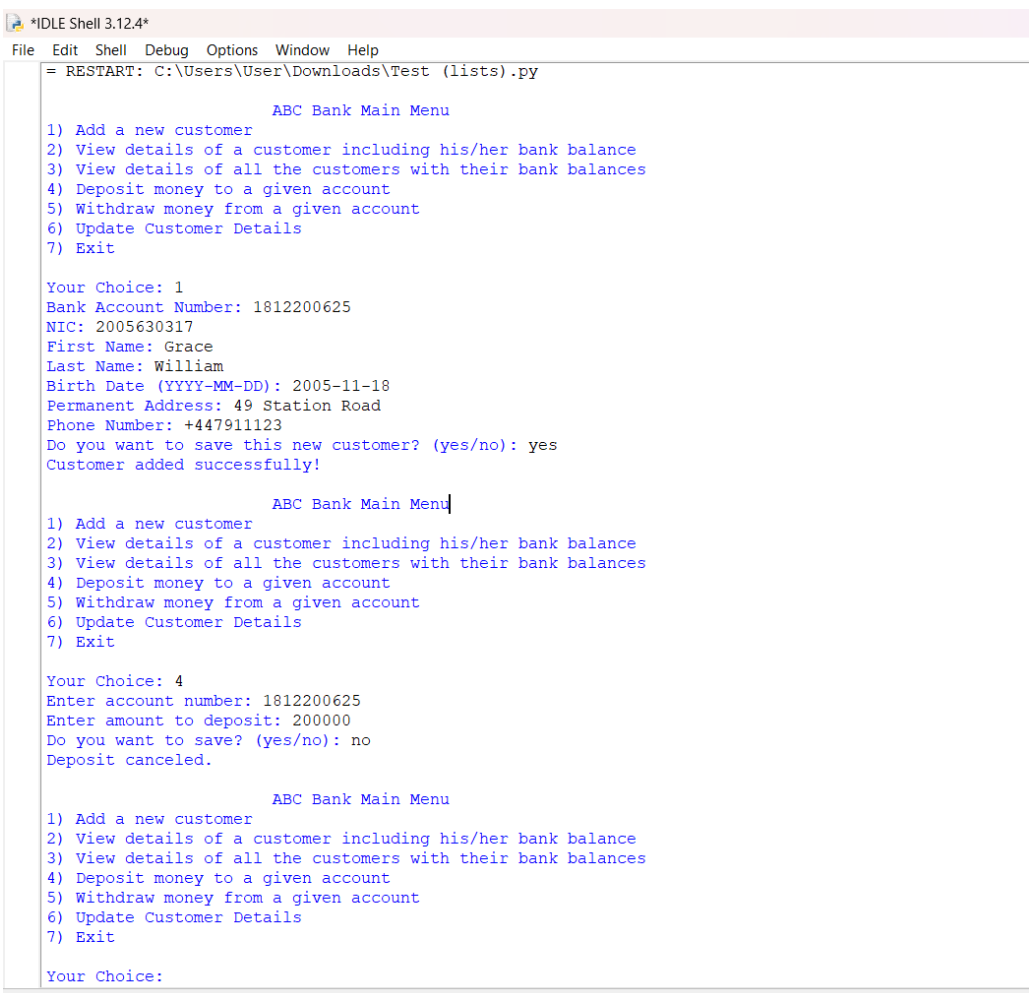


Figure 12: Test Case 9 (Deposit money to a given account-If the user input "no")

1.5 Withdrawal money from a given account

Test Case	Input	Assumed Output	Actual Output	Results
14	1812200625	Enter Account Number: 181220025	Enter Account Number: 181220025	pass
	50000	Enter amount to deposit: 50000	Enter amount to deposit: 50000	Pass
	yes	Do you want to save? (yes/no): yes	Do you want to save? (yes/no): yes	Pass
	-	Withdrawal of 50000.00 successful!	Withdrawal of 50000.00 successful!	Pass
	-	New balance: 150000.00	New balance: 150000.00	Pass

Table 14: Test Case 10 (Withdraw money from a given account)

```
*IDLE Shell 3.12.4*
File  Edit  Shell  Debug  Options  Window  Help
Phone Number: +447911123
Do you want to save this new customer? (yes/no): yes
Customer added successfully!

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 4
Enter account number: 1812200625
Enter amount to deposit: 200000
Do you want to save? (yes/no): yes
Deposit successful!
New balance: 200000.00

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 5
Enter account number: 1812200625
Enter amount to withdraw: 50000
Do you want to save? (yes/no): yes
Withdrawal of 50000.00 successful!
New balance: 150000.00

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: |
```

Figure 13: Test Case 10 (Withdraw money from a given account)

1.5.1 If the user input “no”

Test Case	Input	Assumed Output	Actual Output	Results
15	1812200625	Enter Account Number: 181220025	Enter Account Number: 181220025	pass
	50000	Enter amount to deposit: 50000	Enter amount to deposit: 50000	Pass
	yes	Do you want to save? (yes/no): yes	Do you want to save? (yes/no): yes	Pass
	-	Withdrawal of 50000.00 successful!	Withdrawal of 50000.00 successful!	Pass
	-	New balance: 150000.00	New balance: 150000.00	Pass

Table 15: Test Case 11 (Withdraw money from a given account-If the user input "no")

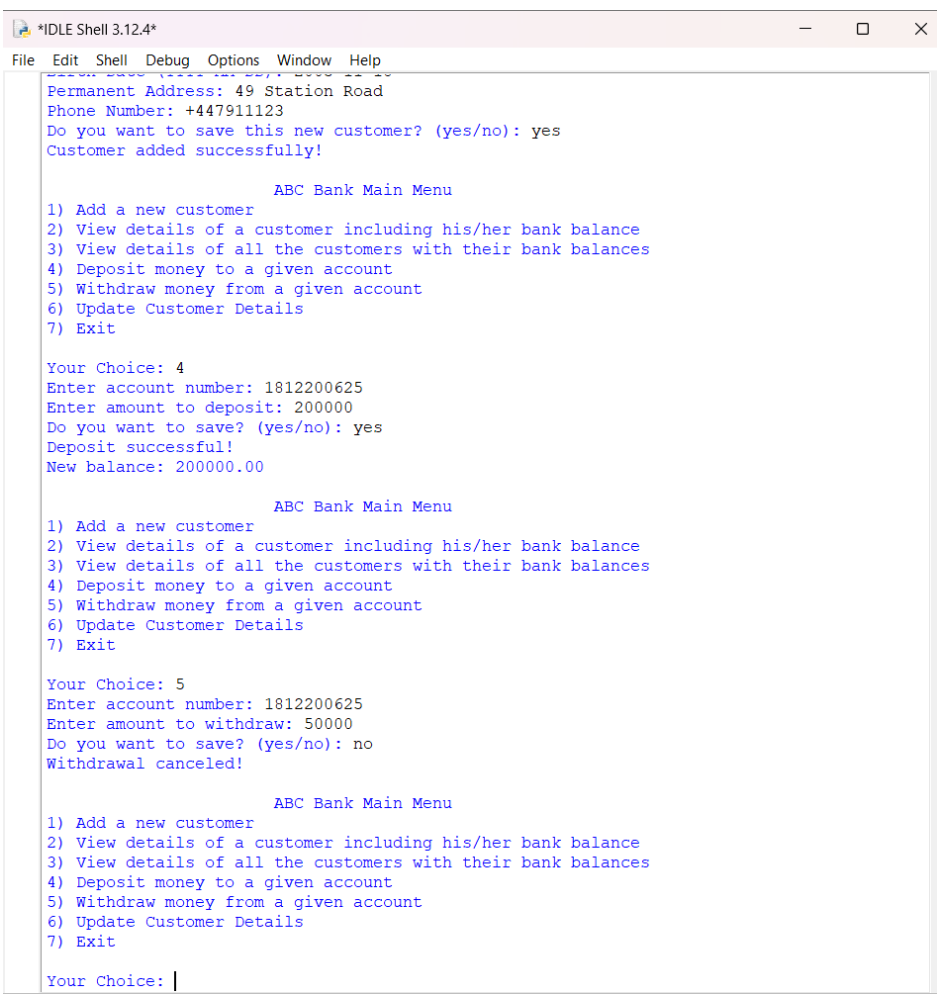


Figure 14: Test Case 11 (Withdraw money from a given account-If the user input "no")

Test Case	Input	Assumed Output	Actual Output	Results
16	1812200625	Enter Account Number: 181220025	Enter Account Number: 181220025	pass
	2110436809	NIC (Leave blank to keep current): 2110436809	NIC (Leave blank to keep current): 2110436809	Pass
		First Name (Leave blank to keep current):	First Name (Leave blank to keep current):	Pass
	-	Last Name (Leave blank to keep current):	Last Name (Leave blank to keep current):	Pass
	-	Birth Date (Leave blank to keep current):	Birth Date (Leave blank to keep current):	Pass
	-	Permanent Address (Leave blank to keep current):	Permanent Address (Leave blank to keep current):	Pass
	-	Phone Number (Leave blank to keep current):	Phone Number (Leave blank to keep current):	Pass
	yes	Do you want to save the updated information? (yes/no): yes	Do you want to save the updated information? (yes/no): yes	Pass
	-	Customer details updated successfully!	Customer details updated successfully!	Pass

Table 16: Test Case 12 (Update Customer Details-NIC)

```
*IDLE Shell 3.12.4*
File  Edit  Shell  Debug  Options  Window  Help
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 1
Bank Account Number: 1812200625
NIC: 2005630317
First Name: Grace
Last Name: William
Birth Date (YYYY-MM-DD): 2005-11-18
Permanent Address: 49 Station Road
Phone Number: +447911123
Do you want to save this new customer? (yes/no): yes
Customer added successfully!

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 6
Enter account number: 1812200625
NIC (Leave blank to keep current): 2110436809
First Name (Leave blank to keep current):
Last Name (Leave blank to keep current):
Birth Date (Leave blank to keep current):
Permanent Address (Leave blank to keep current):
Phone Number (Leave blank to keep current):
Do you want to save the updated information? (yes/no): yes
Customer details updated successfully!

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice:
```

Figure 15: Test Case 12 (Update Customer Details-NIC)

- *First Name*

Test Case	Input	Assumed Output	Actual Output	Results
17	1812200625	Enter Account Number: 181220025	Enter Account Number: 181220025	pass
	-	NIC (Leave blank to keep current):	NIC (Leave blank to keep current):	Pass
	Emma	First Name (Leave blank to keep current): Emma	First Name (Leave blank to keep current): Emma	Pass

	-	Last Name (Leave blank to keep current):	Last Name (Leave blank to keep current):	Pass
	-	Birth Date (Leave blank to keep current):	Birth Date (Leave blank to keep current):	Pass
	-	Permanent Address (Leave blank to keep current):	Permanent Address (Leave blank to keep current):	Pass
	-	Phone Number (Leave blank to keep current):	Phone Number (Leave blank to keep current):	Pass
	yes	Do you want to save the updated information? (yes/no): yes	Do you want to save the updated information? (yes/no): yes	Pass
	-	Customer details updated successfully!	Customer details updated successfully!	Pass

Table 16: Test Case 13 (Update Customer Details-First Name)

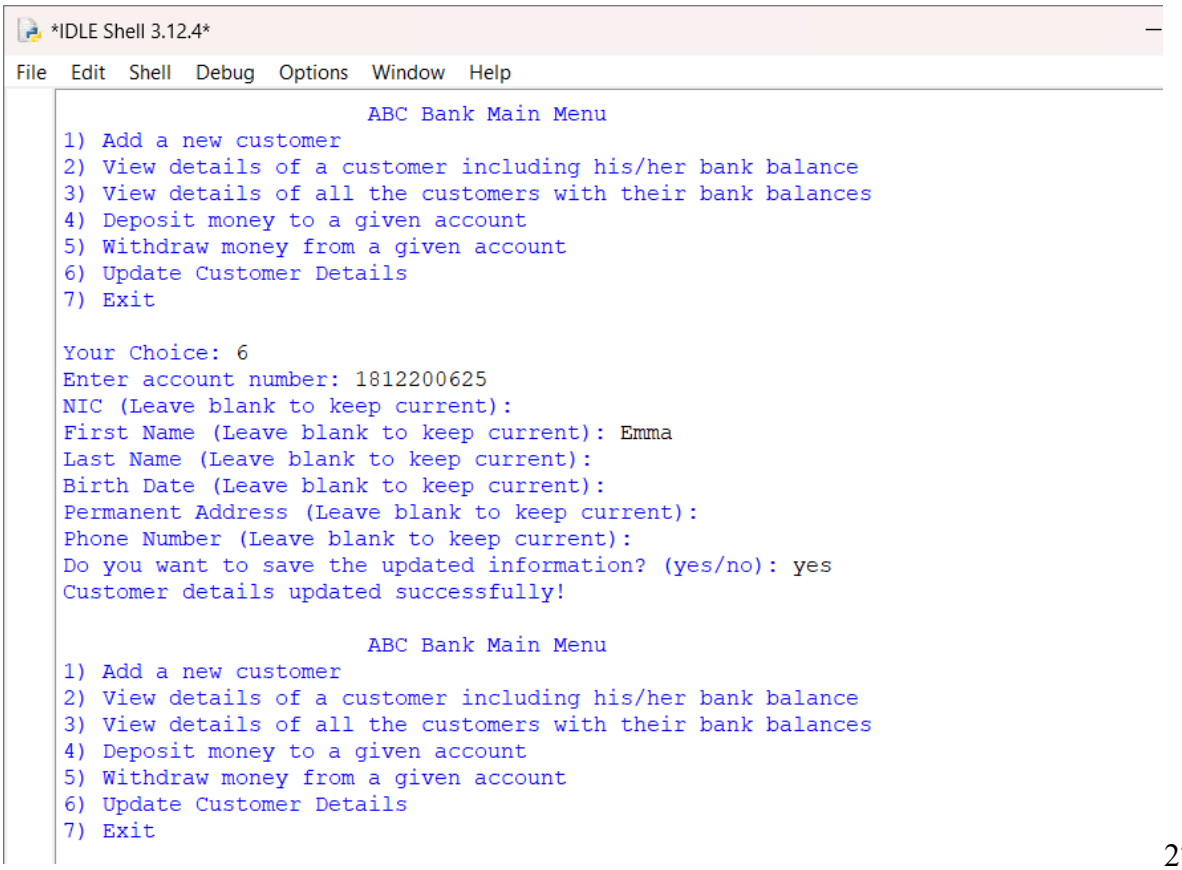


Figure 17: Test Case 13 (Update Customer Details-First Name)

- *Last Name*

Test Case	Input	Assumed Output	Actual Output	Results
18	1812200625	Enter Account Number: 181220025	Enter Account Number: 181220025	pass
	-	NIC (Leave blank to keep current):	NIC (Leave blank to keep current):	Pass
	-	First Name (Leave blank to keep current):	First Name (Leave blank to keep current):	Pass
	Richards	Last Name (Leave blank to keep current): Richards	Last Name (Leave blank to keep current): Richards	Pass
	-	Birth Date (Leave blank to keep current):	Birth Date (Leave blank to keep current):	Pass
	-	Permanent Address (Leave blank to keep current):	Permanent Address (Leave blank to keep current):	Pass
	-	Phone Number (Leave blank to keep current):	Phone Number (Leave blank to keep current):	Pass
	yes	Do you want to save the updated information? (yes/no): yes	Do you want to save the updated information? (yes/no): yes	Pass
	-	Customer details updated successfully!	Customer details updated successfully!	Pass

Table 17:Test Case 14 (Update Customer Details- Last Name)

```
*IDLE Shell 3.12.4*
File  Edit  Shell  Debug  Options  Window  Help
                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 6
Enter account number: 1812200625
NIC (Leave blank to keep current):
First Name (Leave blank to keep current):
Last Name (Leave blank to keep current): Richards
Birth Date (Leave blank to keep current):
Permanent Address (Leave blank to keep current):
Phone Number (Leave blank to keep current):
Do you want to save the updated information? (yes/no): yes
Customer details updated successfully!

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit
```

Figure 18: Test Case 14 (Update Customer Details- Last Name)

- *Birth Date*

Test Case	Input	Assumed Output	Actual Output	Results
19	1812200625	Enter Account Number: 181220025	Enter Account Number: 181220025	pass
	-	NIC (Leave blank to keep current):	NIC (Leave blank to keep current):	Pass
	-	First Name (Leave blank to keep current):	First Name (Leave blank to keep current):	Pass
	-	Last Name (Leave blank to keep current):	Last Name (Leave blank to keep current):	Pass
	2004-05-30	Birth Date (Leave blank to keep current): 2004-05-30	Birth Date (Leave blank to keep current): 2004-05-30	Pass

	-	Permanent Address (Leave blank to keep current):	Permanent Address (Leave blank to keep current):	Pass
	-	Phone Number (Leave blank to keep current):	Phone Number (Leave blank to keep current):	Pass
	Yes	Do you want to save the updated information? (yes/no): yes	Do you want to save the updated information? (yes/no): yes	Pass
	-	Customer details updated successfully!	Customer details updated successfully!	Pass

Table 18: Test Case 15 (Update Customer Details-Birth Date)

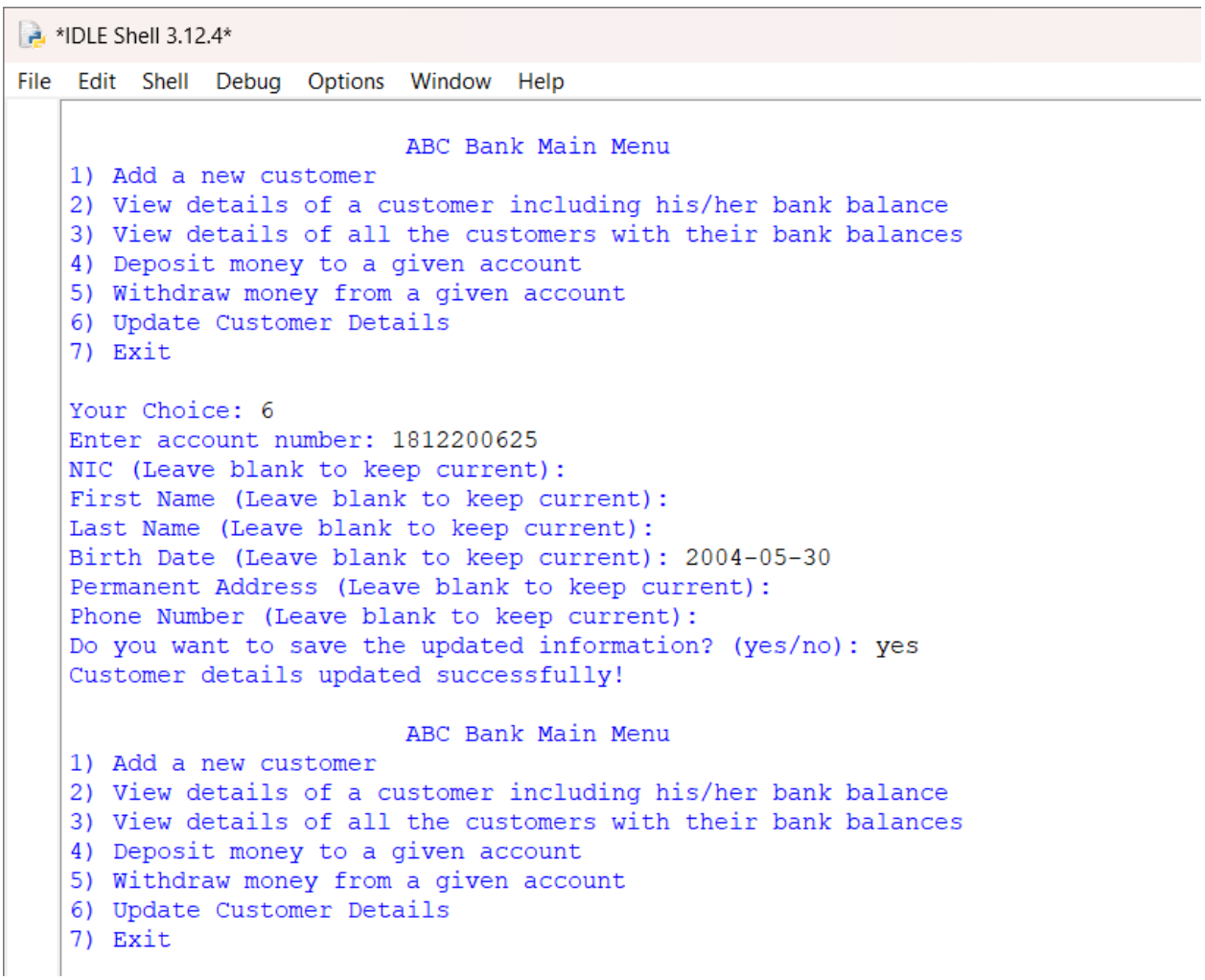


Figure 19: Test Case 15 (Update Customer Details-Birth Date)

- *Permanent Address*

Test Case	Input	Assumed Output	Actual Output	Results
20	1812200625	Enter Account Number: 181220025	Enter Account Number: 181220025	pass
	-	NIC (Leave blank to keep current):	NIC (Leave blank to keep current):	Pass
	-	First Name (Leave blank to keep current):	First Name (Leave blank to keep current):	Pass
	-	Last Name (Leave blank to keep current):	Last Name (Leave blank to keep current):	Pass
	-	Birth Date (Leave blank to keep current):	Birth Date (Leave blank to keep current):	Pass
	11 High Street	Permanent Address (Leave blank to keep current): 11 High Street	Permanent Address (Leave blank to keep current): 11 High Street	Pass
	-	Phone Number (Leave blank to keep current):	Phone Number (Leave blank to keep current):	Pass
	Yes	Do you want to save the updated information? (yes/no): yes	Do you want to save the updated information? (yes/no): yes	Pass
	-	Customer details updated successfully!	Customer details updated successfully!	Pass

Table 19: Test Case 16 (Update Customer Details- Permanent Address)

```
*IDLE Shell 3.12.4*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun  6 2024, 19:30:16) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\User\Downloads\Test (lists).py

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 1
Bank Account Number: 1812200625
NIC: 2005630317
First Name: Grace
Last Name: William
Birth Date (YYYY-MM-DD): 2005-11-18
Permanent Address: 49 Station Road
Phone Number: +447911123
Do you want to save this new customer? (yes/no): yes
Customer added successfully!

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 6
Enter account number: 1812200625
NIC (Leave blank to keep current):
First Name (Leave blank to keep current):
Last Name (Leave blank to keep current):
Birth Date (Leave blank to keep current):
Permanent Address (Leave blank to keep current): 11 High Street
Phone Number (Leave blank to keep current):
Do you want to save the updated information? (yes/no): yes
Customer details updated successfully!
```

Figure 20: Test Case 16 (Update Customer Details- Permanent Address)

- *Phone Number*

Test Case	Input	Assumed Output	Actual Output	Results
21	1812200625	Enter Account Number: 181220025	Enter Account Number: 181220025	pass
	-	NIC (Leave blank to keep current):	NIC (Leave blank to keep current):	Pass
	-	First Name (Leave blank to keep current):	First Name (Leave blank to keep current):	Pass
	-	Last Name (Leave blank to keep current):	Last Name (Leave blank to keep current):	Pass
	-	Birth Date (Leave blank to keep current):	Birth Date (Leave blank to keep current):	Pass
	-	Permanent Address (Leave blank to keep current):	Permanent Address (Leave blank to keep current):	Pass
	+443256591	Phone Number (Leave blank to keep current): +443256591	Phone Number (Leave blank to keep current): +443256591	Pass
	yes	Do you want to save the updated information? (yes/no): yes	Do you want to save the updated information? (yes/no): yes	Pass
	-	Customer details updated successfully!	Customer details updated successfully!	Pass

Table 20: Test Case 17 (Update Customer Details- Phone Number)

```
*IDLE Shell 3.12.4*
File  Edit  Shell  Debug  Options  Window  Help
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 1
Bank Account Number: 1812200625
NIC: 2005630317
First Name: Grace
Last Name: William
Birth Date (YYYY-MM-DD): 2005-11-18
Permanent Address: 49 Station Road
Phone Number: +447911123
Do you want to save this new customer? (yes/no): yes
Customer added successfully!

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 6
Enter account number: 1812200625
NIC (Leave blank to keep current):
First Name (Leave blank to keep current):
Last Name (Leave blank to keep current):
Birth Date (Leave blank to keep current):
Permanent Address (Leave blank to keep current):
Phone Number (Leave blank to keep current): +443256591
Do you want to save the updated information? (yes/no): yes
Customer details updated successfully!

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: |
```

Figure 21:Test Case 17 (Update Customer Details- Phone Number)

1.6 Exit

Test Case	Input	Assumed Output	Actual Output	Results
22	1812200625	Enter Account Number: 181220025	Enter Account Number: 181220025	pass
	200000	Enter amount to deposit: 200000	Enter amount to deposit: 200000	Pass
	yes	Do you want to save? (yes/no): yes	Do you want to save? (yes/no): yes	Pass
	-	Deposit successful!	Deposit successful!	Pass
	-	New balance: 200000.00	New balance: 200000.00	Pass

Table 21: Test Case 18 (Exit)

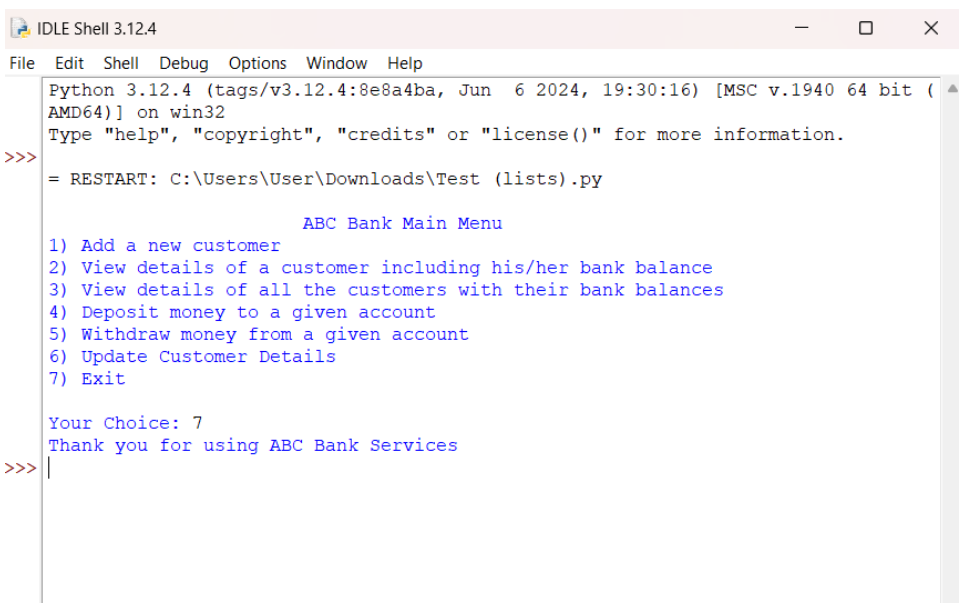


Figure 22: Test Case 18 (Exit)

Negative Cases

Invalid Choice

```
*IDLE Shell 3.12.4*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun  6 2024, 19:30:16) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\User\Downloads\Test (lists).py

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 12
Invalid choice! Please try again.

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: |
```

Figure 23: Test Case 19 (Negative Case-Invalid Choice)

Boundary Conditions

```
*IDLE Shell 3.12.4*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun  6 2024, 19:30:16) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\User\Downloads\Test (lists).py

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 1
Bank Account Number: 1812200625
NIC: 2005446282
First Name: Grace
Last Name: William
Birth Date (YYYY-MM-DD): 2005-11-18
Permanent Address: 49 Station Road
Phone Number: +447911123
Do you want to save this new customer? (yes/no): yes
Customer added successfully!

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 4
Enter account number: 1812200625
Enter amount to deposit: 100000000
Deposit amount exceeds maximum limit of 5,000,000. Please enter a smaller amount.
Enter amount to deposit:
```

Figure 24: Test Case 20 (Negative Case-Condition Boundary)

```
*IDLE Shell 3.12.4*
File  Edit  Shell  Debug  Options  Window  Help
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 1
Bank Account Number: 1812200625
NIC: 2005646331
First Name: Grace
Last Name: William
Birth Date (YYYY-MM-DD): 2005-11-18
Permanent Address: 49 Station Road
Phone Number: +447911123
Do you want to save this new customer? (yes/no): yes
Customer added successfully!

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 4
Enter account number: 1812200625
Enter amount to deposit: 200000
Do you want to save? (yes/no): yes
Deposit successful!
New balance: 200000.00

                        ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 5
Enter account number: 1812200625
Enter amount to withdraw: 200000
Withdrawal amount exceeds maximum limit of 60,000. Please enter a smaller amount.
Enter amount to withdraw:
```

Figure 25: Test Case 21 (Negative Case-Condition Boundary)

```
*IDLE Shell 3.12.4*
File  Edit  Shell  Debug  Options  Window  Help
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 1
Bank Account Number: 1812200625
NIC: 2005646331
First Name: Grace
Last Name: William
Birth Date (YYYY-MM-DD): 2005-11-18
Permanent Address: 49 Station Road
Phone Number: +447911123
Do you want to save this new customer? (yes/no): yes
Customer added successfully!

                      ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 4
Enter account number: 1812200625
Enter amount to deposit: 5000000
Do you want to save? (yes/no): yes
Deposit successful!
New balance: 5000000.00

                      ABC Bank Main Menu
1) Add a new customer
2) View details of a customer including his/her bank balance
3) View details of all the customers with their bank balances
4) Deposit money to a given account
5) Withdraw money from a given account
6) Update Customer Details
7) Exit

Your Choice: 5
Enter account number: 1812200625
Enter amount to withdraw: 50
Withdrawal amount must exceed 100.Please enter a higher number.
Enter amount to withdraw:
```

Figure 26: Test Case 22 (Negative Case-Condition Boundary)